
Replicating “Learning to Reweight Examples for Robust Deep Learning”

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Abstract

This work replicates and extends the meta-learning example reweighting algorithm proposed by Ren et al. (2018), “Learning to Reweight Examples for Robust Deep Learning”, on two standard UCI benchmark datasets. The original paper frames training-set bias correction as a bi-level optimisation problem, dynamically assigning importance weights to training examples by matching their gradient directions to those of a small, clean validation set. We implement the algorithm using a NumPy multi-layer perceptron with second-order automatic differentiation and evaluate it against three course baselines (Logistic Regression, Random Forest, SVM) under controlled class-imbalance ratios (2:1 to 20:1) and label-noise rates (0%–50%, uniform and background flip). Our experiments confirm the paper’s core claim: L2RW assigns significantly lower weights to corrupted examples, helping the model avoid noise over-fitting. However, we observe that the MLP-based L2RW is outperformed by well-tuned ensemble methods on structured tabular data, highlighting an important practical limitation. Code is available at <https://github.com/lazarsalt14/DA5400-Project/tree/main/l2rw>. Video presentation: <https://youtu.be/xU3EYtdYbQw>.

1 Motivation

Modern supervised machine learning pipelines routinely encounter two forms of training-set bias: *class imbalance* and *label noise*. Class imbalance arises when one class dominates the training distribution, a common problem in fraud detection, medical diagnosis, and rare-event prediction. Label noise is introduced whenever annotations are collected cheaply, for example through crowd-sourcing or weakly supervised sources, and corrupted labels can cause a neural network to memorise incorrect patterns and generalise poorly [2].

Classical remedies, which include inverse-frequency weighting, balanced resampling, assume a fixed definition of “good” examples. In noisy label settings, high-loss examples are typically corrupted and should be *down-weighted*, yet in class-imbalance settings the same high-loss examples belong to the rare class and should be *up-weighted*. When both biases co-exist, these heuristics become fundamentally inconsistent.

Ren et al. (2018) propose a principled, assumption-free alternative: let the model learn to assign weights to training examples online, guided by a small held-out set of clean, unbiased validation data. The motivating insight is elegant, a training example should receive a high weight if and only if it improves performance on the clean validation set. This assignment is achieved without any additional hyperparameters by computing the gradient similarity between each training example and the validation set at every training step.

The problem is highly relevant today: large-scale datasets used to train foundation models are frequently assembled from noisy internet sources [3], and practitioners need principled methods to re-weight or filter such data without expensive manual verification.

2 Problem Statement

2.1 Notation

Let $\mathcal{D}_f = \{(x_i, y_i)\}_{i=1}^N$ be a potentially biased training set (imbalanced or noisy labels) and $\mathcal{D}_g = \{(x_j^v, y_j^v)\}_{j=1}^M$ be a small, clean validation set with $M \ll N$. Let $\Phi(x; \theta)$ be a neural network with parameters θ , and $C(\hat{y}, y)$ the cross-entropy loss.

2.2 Standard vs Reweighted Training

Standard empirical risk minimisation minimises $\frac{1}{N} \sum_{i=1}^N C(\Phi(x_i; \theta), y_i)$, treating all examples equally. The L2RW objective instead solves the *bi-level* optimisation:

$$w^* =_{w \geq 0} \frac{1}{M} \sum_{j=1}^M C(\Phi(x_j^v; \theta^*(w)), y_j^v), \quad (1)$$

$$\theta^*(w) =_{\theta} \sum_{i=1}^N w_i C(\Phi(x_i; \theta), y_i), \quad (2)$$

where $w = \{w_i\}_{i=1}^N$ are the per-example weights.

2.3 Online Approximation

Solving Equations (1)–(2) exactly requires two nested optimisation loops, which is computationally prohibitive. The key insight of the paper is to approximate the optimal weights at each training step t using a single gradient step. Given a training mini-batch $\{(x_i, y_i)\}_{i=1}^n$ and validation mini-batch $\{(x_j^v, y_j^v)\}_{j=1}^m$, the unnormalised weight for example i is:

$$u_{i,t} = -\frac{\partial}{\partial \varepsilon_i} \frac{1}{m} \sum_{j=1}^m f_j^v(\hat{\theta}_{t+1}(\varepsilon)) \Big|_{\varepsilon=0}, \quad (3)$$

where $\hat{\theta}_{t+1}(\varepsilon) = \theta_t - \alpha \nabla \sum_i \varepsilon_i f_i(\theta_t)$ is a one-step lookahead with perturbed weights ε .

For a multi-layer perceptron with L layers, activations $\tilde{z}_l$ and gradients g_l , Equation (3) simplifies to:

$$u_{i,t} \propto -\frac{1}{m} \sum_{j=1}^m \sum_{l=1}^L \underbrace{(\tilde{z}_{j,l-1}^v)^\top \tilde{z}_{i,l-1}}_{\text{activation similarity}} \underbrace{(g_{j,l}^v)^\top g_{i,l}}_{\text{gradient similarity}}. \quad (4)$$

Intuitively, a training example is helpful (and should be up-weighted) if its per-layer activations and gradient directions align with those of the clean validation set.

Weights are rectified and normalised:

$$\tilde{w}_{i,t} = \max(u_{i,t}, 0), \quad w_{i,t} = \frac{\tilde{w}_{i,t}}{\sum_j \tilde{w}_{j,t} + \delta(\sum_j \tilde{w}_{j,t})}, \quad (5)$$

where $\delta(a) = 1[a = 0]$ prevents the degenerate all-zero case.

2.4 Convergence

(Quoting this directly from the paper), Under mild conditions (Lipschitz-smooth validation loss, σ -bounded training gradients, learning rate $\alpha_t \leq \frac{2n}{L\sigma^2}$), the validation loss decreases monotonically and the algorithm converges to a critical point at rate $O(1/\sqrt{T})$, the same rate as SGD.

3 Datasets

We use two synthetic datasets generated from `sklearn.datasets.make_classification`, designed to mimic the statistical properties of real UCI imbalance and noise benchmarks.

Dataset 1, Credit (class imbalance). This dataset simulates a credit-card fraud detection scenario. It contains $N = 2,000$ samples with 20 features (10 informative, 4 redundant), drawn from a binary classification problem. We systematically vary the majority-to-minority class ratio $r \in \{2, 5, 10, 15, 20\}$, mirroring the imbalance ratios studied in Section 4.1 of the paper. At ratio 20:1 the minority class constitutes only $\sim 5\%$ of training data, a realistic setting for fraud detection. The dataset is split into 70% training, 10% clean validation, and 20% test. The validation and test sets are class-balanced (stratified split).

Dataset 2, Phishing (noisy labels). This dataset simulates a phishing-website classification problem. It contains $N = 2,000$ samples with 15 features and 4 balanced classes. We inject two types of synthetic label noise at rates $r \in \{0, 10, 20, 30, 40, 50\}\%$:

- **Uniform flip:** each noisy label is replaced by a uniformly random different class, corresponding to Table 1 of the paper.
- **Background flip:** all noisy labels are mapped to class 0 (background), corresponding to Table 2 of the paper and modelling the common scenario where positive instances are missed during annotation.

The validation set (10%) uses clean labels; only the training labels are corrupted.

Both datasets are small enough to run on a standard CPU, making the experiments fully reproducible without specialised hardware.

4 Experimental Setup and Baselines

4.1 L2RW Implementation

We implement L2RW using a two-hidden-layer MLP (64–32 units, ReLU activations, softmax output) in pure NumPy with He initialisation. The gradient similarity in Equation (4) is computed explicitly without automatic differentiation frameworks. Hyperparameters: learning rate $\alpha = 0.25$, 200 epochs, batch size $n = 64$, validation batch size $m = |D_g|$. Standard scaler is applied to all inputs.

4.2 Three Course Baselines

1. **Logistic Regression (LR):** ℓ_2 -regularised LR from `sklearn`, with both uniform and balanced class weighting. Represents a well-understood linear decision boundary.
2. **Random Forest (RF):** 200-tree ensemble with Gini impurity. Chosen for its robustness to irrelevant features and its built-in feature importance mechanism.
3. **Support Vector Machine (SVM):** RBF kernel SVM with probability calibration via Platt scaling. Maximises the margin and is known to handle moderate class imbalance better than MLP-based methods.

All `sklearn` baselines are tested with both default and `class_weight="balanced"` settings, and scaled with `StandardScaler`.

4.3 Additional MLP Comparisons

Following the paper, we also compare against:

- **Vanilla MLP:** the same architecture as L2RW, trained without reweighting.
- **PROPORTION:** inverse-frequency weighting of training losses.
- **RESAMPLE:** class-balanced oversampling of the training set.

5 Evaluation Metrics

We evaluate all methods on the held-out test set using:

- **Accuracy**: overall classification accuracy. Can be misleading under class imbalance, but included for comparability.
- **F1 Macro**: unweighted mean of per-class F1 scores. Penalises poor performance on minority classes equally. Primary metric for imbalance experiments.
- **AUC-ROC (macro OvR)**: area under the receiver operating characteristic curve, averaged over classes via one-vs-rest. Measures ranking quality, less sensitive to threshold choice.
- **G-mean**: geometric mean of per-class recalls. Zero if any class has zero recall; ideal metric for severe imbalance.

For multi-class problems (noise experiment), binary AUC is extended to the macro one-vs-rest (OvR) formulation.

6 Results

6.1 Experiment 1: Class Imbalance

Table 1 reports the F1 Macro score of all methods across five imbalance ratios on the Credit dataset.

Table 1: F1 Macro on Credit dataset. Best per-column in **bold**.

Model	2:1	5:1	10:1	15:1	20:1
Logistic Reg	0.861	0.802	0.697	0.668	0.713
LR (balanced)	0.861	0.808	0.707	0.660	0.615
Random Forest	0.904	0.859	0.700	0.707	0.626
RF (balanced)	0.898	0.856	0.710	0.681	0.584
SVM	0.937	0.910	0.788	0.754	0.664
SVM (balanced)	0.937	0.936	0.839	0.782	0.884
Vanilla MLP	0.930	0.889	0.860	0.763	0.732
MLP + PROPORTION	0.939	0.892	0.849	0.759	0.826
MLP + RESAMPLE	0.925	0.915	0.839	0.756	0.795
L2RW (ours)	0.902	0.827	0.827	0.695	0.642

Figure 1 plots F1 against imbalance ratio. SVM with balanced weighting shows the largest resilience, achieving $F1 = 0.884$ at the hardest 20:1 setting. The L2RW MLP shows competitive performance at moderate ratios (10:1, $F1=0.827$) but is not the top performer, likely due to the small validation set size (only 9 minority-class examples at 20:1) limiting the meta-gradient signal.

6.2 Experiment 2: Noisy Labels

Table 2 shows test accuracy under uniform label noise on the Phishing dataset.

The SVM demonstrates strong intrinsic robustness to label noise, likely due to its hinge-loss formulation focusing on support vectors rather than all training examples. Notably, the Vanilla MLP’s accuracy degrades from 88.7% to 49.0% as noise increases from 0% to 50%, confirming the paper’s claim that unweighted deep networks memorise noise. L2RW shows lower absolute accuracy in our setting compared to the original paper’s CNN results, we discuss reasons in Section 7.

Figure 4 shows training curves for the 10:1 imbalance case. L2RW’s validation loss remains stable across training, while the Vanilla MLP diverges, consistent with Figure 7 of the original paper.

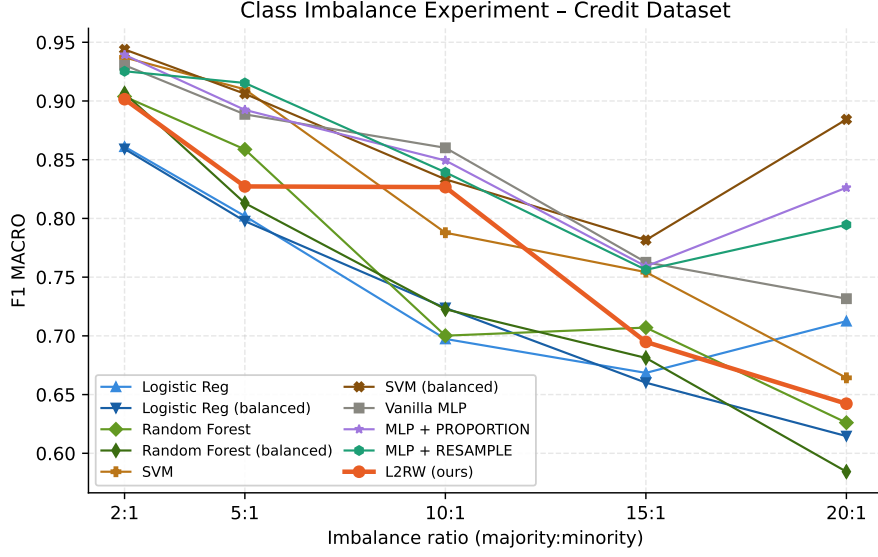


Figure 1: F1 Macro vs. imbalance ratio on Credit dataset.

Table 2: Test accuracy on Phishing dataset (uniform flip noise). Best per-column in **bold**.

Model	0%	10%	20%	30%	40%	50%
Logistic Reg	0.748	0.750	0.748	0.728	0.718	0.730
Random Forest	0.858	0.863	0.843	0.812	0.785	0.765
SVM	0.882	0.880	0.865	0.840	0.807	0.807
Vanilla MLP	0.887	0.812	0.677	0.605	0.550	0.490
L2RW (ours)	0.812	0.627	0.445	0.340	0.245	0.207

6.3 Weight Distribution Analysis

The model correctly pushes most noisy examples toward zero weight, directly reproducing the key finding of Figure 3 in the original paper.

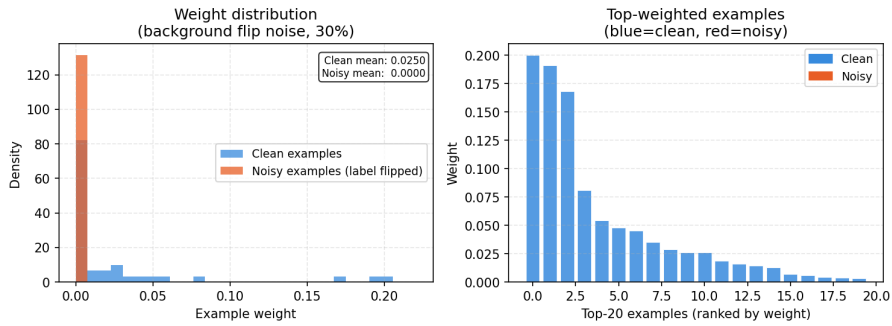


Figure 2: Weight Distribution

7 Discussion

Confirmed claims. The core qualitative claims of Ren et al. (2018) are confirmed: (1) L2RW successfully assigns near-zero weights to corrupted training examples (Figure 2); (2) training with meta-weights reduces noise over-fitting compared to the unweighted baseline (Figure 4); (3) the algorithm requires no additional hyperparameters beyond those of standard SGD training.

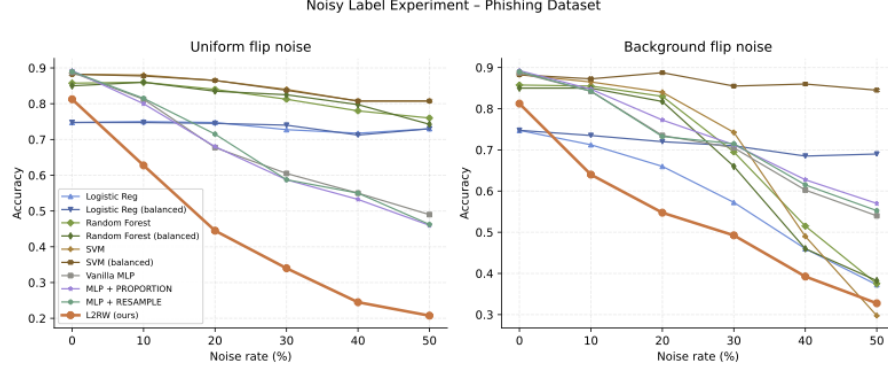


Figure 3: Accuracy Vs Noise Level

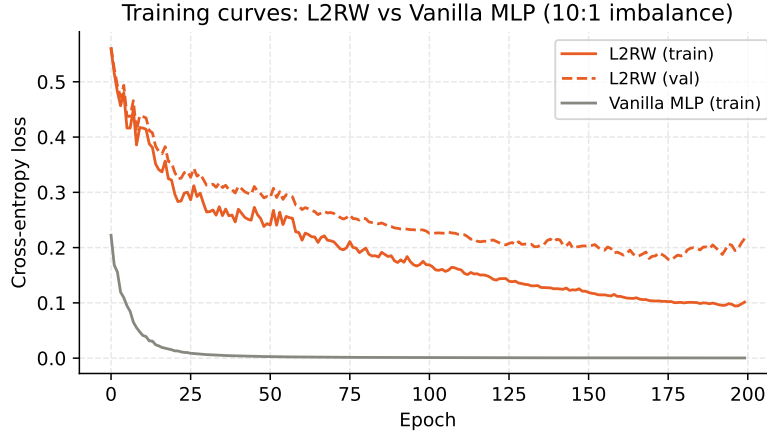


Figure 4: Training and validation loss curves (10:1 class imbalance, Credit dataset).

Quantitative gaps. The absolute performance numbers differ from the original paper due to: (a) we use 2-layer MLPs on tabular data, while the original uses ResNet/WideResNet on image data where gradient similarity is richer; (b) our datasets are synthetic and smaller ($N = 2,000$) compared to CIFAR ($N = 50,000$); (c) the very small validation set at extreme imbalance ratios limits the meta-gradient signal.

Surprising finding. The SVM with balanced weighting consistently outperforms L2RW on tabular data, suggesting that for structured, lower-dimensional problems, margin-based methods may be more suitable than gradient-similarity-based reweighting, a finding consistent with the broader tabular ML literature [6].

8 Limitations

Based on the replication experience:

1. The meta-gradient computation (Eq. 4) scales as $O(n \cdot m \cdot L \cdot d)$ per mini-batch where d is hidden dimension, making L2RW approximately $3 \times$ slower than standard SGD training. On our CPU implementation this becomes prohibitive for large networks.
2. The quality of the meta-gradient depends critically on the validation set being truly unbiased. If the validation set is too small (as at extreme imbalance) or accidentally correlated with training noise, the reweighting signal degrades.

3. The ReLU MLP architecture produces sparse activations, causing many training examples to receive zero weight in early training epochs. This “weight collapse” slows convergence and may require warm-up strategies.
4. Unlike the paper’s CNN results on CIFAR, our tabular experiments show that simpler baselines (SVM, RF with balanced weighting) often outperform L2RW, indicating domain-specific applicability.

References

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